# Sanskruti Sarees Data Dictionary and ERD Note

This document captures the finalized database design for Sanskruti Sarees, including the core entity relationships, field meanings, business-rule constraints, and implementation notes needed before backend development begins.[cite:57][cite:61][cite:73]

### Overview

The schema is designed for a premium saree and chaniya choli e-commerce website with online-only payments, dynamic product filters, weight-based shipping, review eligibility based on delivered orders, and device-upload-only media management.[cite:73][cite:79]

The model uses normalized lookup tables and relationship tables to avoid hardcoding future product attributes into the `products` table, which keeps the design cleaner and easier to extend over time.[cite:57][cite:61][cite:79]

### ERD Note

### Core relationships

- One customer can have many addresses, orders, wishlist items, cart items, cancellation requests, and reviews.
- One order has many order items.
- One product belongs to one category, and may optionally point to one fabric, one collection, and one color.
- One product can have many media files and many dynamic filter assignments.
- One filter type has many filter values; one product can have many filter values through `product_filter_values`.
- One approved cancellation can create a review block for a customer-product pair, preventing only future review creation.[cite:57][cite:61][cite:73]

### Main relationship map

```
users
 ├── addresses
 ├── orders
 │    └── order_items
 │         ├── reviews
 │         └── cancellation_requests
 ├── wishlists
 ├── cart_items
 └── review_blocks

products
```

```
  ├── product_media
  ├── order_items
  ├── reviews
  ├── review_blocks
  └── product_filter_values ── filter_values ── filter_types

categories ── products
fabrics ── products
collections ── products
colors ── products
states ── addresses
shipping_rates ── orders (used by checkout logic)
site_settings ── global business rules
admin_users ── product_media uploads / cancellation resolution
```

## Data dictionary

`users`

| Column | Meaning |
|---|---|
| `user_id` | Unique customer account ID |
| `email_address` | Customer email for login and communication |
| `phone_number` | Customer phone number |
| `password_hash` | Hashed password, never plain text |
| `full_name` | Customer full name |
| `is_verified` | Whether account contact verification is completed |
| `is_active` | Whether account can currently log in and place orders |
| `created_at` | Account creation timestamp |
| `updated_at` | Last profile/account update timestamp |

`addresses`

| Column | Meaning |
|---|---|
| `address_id` | Unique address row ID |
| `customer_user_id` | Which user owns this address |
| `recipient_full_name` | Delivery recipient name |
| `recipient_phone_number` | Delivery recipient phone |
| `address_line_1` | Primary address line |
| `address_line_2` | Optional secondary address line |
| `city_name` | City name |

| Column | Meaning |
| --- | --- |
| `state_id` | Canonical state reference used for reliable shipping logic |
| `postal_code` | PIN or ZIP code |
| `country_code` | Country code, such as `IN` |
| `is_default_address` | Whether this is the customer's default address |
| `created_at` | Address creation timestamp |
| `updated_at` | Last address update timestamp |

`states`

| Column | Meaning |
| --- | --- |
| `state_id` | Unique state/region ID |
| `country_code` | Country the state belongs to |
| `state_name` | Canonical state name, such as `Gujarat` |
| `state_code` | Short state code, such as `GJ` |
| `local_name` | Optional local-language state name |
| `is_active` | Whether this state can be selected |
| `created_at` | Creation timestamp |
| `updated_at` | Update timestamp |

`categories`

| Column | Meaning |
| --- | --- |
| `category_id` | Unique category ID |
| `category_name` | Category name, such as `Saree` or `Chaniya Choli` |
| `category_slug` | URL-safe category identifier |
| `is_active` | Whether category is active |

`fabrics`

| Column | Meaning |
| --- | --- |
| `fabric_id` | Unique fabric ID |
| `fabric_name` | Fabric name |
| `fabric_slug` | URL-safe fabric identifier |
| `cover_media_file_key` | Storage key for homepage/listing fabric image |

| Column | Meaning |
| --- | --- |
| `display_order` | Sort order for admin and homepage |
| `show_on_homepage` | Whether this fabric card appears on homepage |
| `is_active` | Whether fabric is active |
| `created_at` | Creation timestamp |
| `updated_at` | Update timestamp |

`collections`

| Column | Meaning |
| --- | --- |
| `collection_id` | Unique collection ID |
| `collection_name` | Collection name, such as `Wedding` or `Festive` |
| `collection_slug` | URL-safe collection identifier |
| `cover_media_file_key` | Storage key for collection card image |
| `display_order` | Sort order |
| `show_on_homepage` | Whether this collection card appears on homepage |
| `is_active` | Whether collection is active |
| `created_at` | Creation timestamp |
| `updated_at` | Update timestamp |

`colors`

| Column | Meaning |
| --- | --- |
| `color_id` | Unique color ID |
| `color_name` | Color display name |
| `color_hex_code` | Hex shade used for swatch display |
| `display_order` | Sort order |
| `is_active` | Whether color is active |
| `created_at` | Creation timestamp |
| `updated_at` | Update timestamp |

`filter_types`

| Column | Meaning |
|---|---|
| `filter_type_id` | Unique filter type ID |
| `filter_type_name` | Admin/display name, such as `Work Type` |
| `filter_type_slug` | URL-safe identifier |
| `filter_input_type` | UI control type: dropdown, multi-select, checkbox group |
| `applies_to_category_id` | Optional category restriction; NULL means all supported categories |
| `show_in_listing_filters` | Whether this filter appears in frontend listing filter UI |
| `show_in_product_details` | Whether selected values appear on product detail page |
| `display_order` | Sort order |
| `is_active` | Whether filter type is active |
| `created_at` | Creation timestamp |
| `updated_at` | Update timestamp |

`filter_values`

| Column | Meaning |
|---|---|
| `filter_value_id` | Unique filter value ID |
| `filter_type_id` | Parent filter type |
| `filter_value_name` | Option name, such as `Zari` |
| `filter_value_slug` | Normalized identifier, such as `zari` |
| `display_order` | Sort order within the filter |
| `is_active` | Whether value is active |
| `created_at` | Creation timestamp |
| `updated_at` | Update timestamp |

**Constraint:** `UNIQUE (filter_type_id, filter_value_slug)` prevents duplicate options such as two `zari` values under the same filter type.[cite:57][cite:61]

`products`

| Column | Meaning |
|---|---|
| `product_id` | Unique product ID |
| `product_category_id` | Category reference |
| `fabric_id` | Primary fabric reference |

| Column | Meaning |
|---|---|
| collection_id | Primary collection reference |
| color_id | Primary color reference |
| product_name | Product display name |
| product_slug | URL-safe product identifier |
| product_sku | Internal stock code |
| product_description | Product description |
| product_care_instructions | Care and washing text |
| selling_price_inr | Current selling price in INR |
| original_price_inr | Optional original/strikethrough price |
| available_stock_quantity | Current stock count |
| weight_in_grams | Product weight used for shipping calculation |
| is_active | Whether product is published |
| is_premium_product | Manual premium flag |
| is_saree_normal | Saree tag: Normal |
| has_ready_blouse | Saree tag: With Ready Blouse |
| is_bridal_chaniya_choli | CC tag: Bridal |
| is_sider_chaniya_choli | CC tag: Sider |
| seo_meta_title | SEO title |
| seo_meta_description | SEO description |
| created_at | Creation timestamp |
| updated_at | Update timestamp |

`product_filter_values`

| Column | Meaning |
|---|---|
| product_id | Product receiving the filter assignment |
| filter_value_id | Filter value assigned to that product |
| created_at | Assignment timestamp |

This table is the only place where dynamic filter selections are stored for products, so no new filter columns ever need to be added to `products`.[cite:73][cite:79]

`product_media`

| Column | Meaning |
|---|---|
| `product_media_id` | Unique media row ID |
| `product_id` | Product this media belongs to |
| `storage_provider_name` | Storage backend name |
| `storage_file_key` | File path/object key in storage |
| `uploaded_file_name` | Original file name from admin device |
| `uploaded_file_mime_type` | MIME type of uploaded file |
| `uploaded_file_size_in_bytes` | File size |
| `media_type` | `image`, `video`, or `gif` |
| `media_width_px` | Width in pixels when available |
| `media_height_px` | Height in pixels when available |
| `video_duration_in_seconds` | Video duration when applicable |
| `display_order` | Ordering in gallery |
| `is_cover_media` | Whether this media is the main product cover |
| `uploaded_by_admin_user_id` | Admin who uploaded the file |
| `created_at` | Upload timestamp |

Application rules: maximum 9 images per product, maximum 3 videos/GIFs per product, and exactly one cover media per product enforced in the service layer before insert/update.[cite:41][cite:47][cite:73]

`cart_items`

| Column | Meaning |
|---|---|
| `cart_item_id` | Unique cart row ID |
| `customer_user_id` | Logged-in user owning the cart row, nullable for guest flow |
| `guest_session_id` | Guest session identifier when user is not logged in |
| `product_id` | Product added to cart |
| `quantity` | Quantity in cart |
| `created_at` | Added timestamp |
| `updated_at` | Last cart update timestamp |

`wishlists`

| Column | Meaning |
|---|---|
| `wishlist_item_id` | Unique wishlist row ID |
| `customer_user_id` | User who saved the product |
| `product_id` | Wishlisted product |
| `created_at` | Added timestamp |

**Constraint:** `UNIQUE (customer_user_id, product_id)` prevents duplicate wishlist entries.

`orders`

| Column | Meaning |
|---|---|
| `order_id` | Unique order ID |
| `order_number` | Human-friendly order number |
| `customer_user_id` | Customer who placed the order |
| `shipping_address_id` | Selected address reference |
| `shipping_address_snapshot_json` | Immutable copy of address at order time |
| `order_status` | Order state: placed, processing, shipped, delivered, cancelled |
| `payment_status` | Payment state: pending, paid, failed, refunded |
| `order_subtotal_inr` | Sum of item totals before shipping |
| `shipping_zone` | `gujarat`, `india`, or `international` |
| `shipping_charge_inr` | Final computed shipping amount |
| `order_grand_total_inr` | Final payable amount |
| `razorpay_order_id` | Razorpay order reference |
| `razorpay_payment_id` | Razorpay payment reference |
| `internal_admin_notes` | Admin-only notes |
| `delivered_at` | Delivery timestamp used for cancellation window logic |
| `created_at` | Order creation timestamp |
| `updated_at` | Last order update timestamp |

Shipping logic uses normalized address data: Gujarat orders get zero shipping, while India and international orders use active weight-based rates with minimum charges.[cite:43][cite:46][cite:49]

`order_items`

| Column | Meaning |
|---|---|
| `order_item_id` | Unique order line ID |
| `order_id` | Parent order |
| `product_id` | Purchased product |
| `product_snapshot_json` | Immutable product snapshot at purchase time |
| `ordered_quantity` | Quantity purchased |
| `unit_selling_price_inr` | Price per unit at purchase time |
| `line_total_price_inr` | Quantity multiplied by unit price |
| `chaniya_choli_waist_size` | Customer-entered waist size for CC orders only |
| `chaniya_choli_length` | Customer-entered length for CC orders only |
| `created_at` | Creation timestamp |

Only these two Chaniya Choli measurement fields exist, and they live at order-item level because they can vary per order.[cite:73][cite:79]

`shipping_rates`

| Column | Meaning |
|---|---|
| `shipping_rate_id` | Unique shipping rate row ID |
| `shipping_zone` | Supported zone: `india` or `international` |
| `rate_per_gram_inr` | Weight-based shipping amount per gram |
| `minimum_shipping_charge_inr` | Minimum shipping charge for the zone |
| `is_active` | Whether this rate row is currently active |
| `created_at` | Creation timestamp |
| `updated_at` | Update timestamp |

`cancellation_requests`

| Column | Meaning |
|---|---|
| `cancellation_request_id` | Unique cancellation request ID |
| `order_id` | Related order |
| `order_item_id` | Related order item if tracked at item level |
| `customer_user_id` | Customer who requested cancellation |
| `request_source` | Source of request, currently WhatsApp |

| Column | Meaning |
|---|---|
| `cancellation_status` | `pending`, `approved`, or `rejected` |
| `customer_request_note` | Optional note captured from customer message |
| `admin_decision_note` | Admin note/reason |
| `requested_at` | Request timestamp |
| `resolved_at` | Resolution timestamp |
| `resolved_by_admin_user_id` | Admin who approved/rejected the request |

The request window is valid only within 48 hours after `orders.delivered_at`, and only admin can finalize approval or rejection.[cite:73][cite:79]

`reviews`

| Column | Meaning |
|---|---|
| `review_id` | Unique review ID |
| `product_id` | Reviewed product |
| `customer_user_id` | Customer who wrote the review |
| `order_item_id` | Delivered purchased line item proving eligibility |
| `rating_value` | Numeric rating from 1 to 5 |
| `review_title` | Optional review heading |
| `review_text` | Review body |
| `is_visible` | Whether review is publicly visible |
| `created_at` | Review creation timestamp |
| `updated_at` | Last review update timestamp |

**Constraint:** `UNIQUE (customer_user_id, product_id)` allows at most one review per customer per product.[cite:57][cite:61]

`review_blocks`

| Column | Meaning |
|---|---|
| `review_block_id` | Unique review block row ID |
| `customer_user_id` | Customer blocked from future review creation |
| `product_id` | Product on which new review is blocked |
| `order_item_id` | Order item tied to the approved cancellation |
| `cancellation_request_id` | Cancellation row that triggered the block |

| Column | Meaning |
| --- | --- |
| `block_reason` | Reason for blocking, such as `approved_cancellation` |
| `created_at` | Block creation timestamp |

**Constraint:** `UNIQUE (customer_user_id, product_id)` prevents duplicate block rows for the same customer-product pair.[cite:57][cite:61]

`admin_users`

| Column | Meaning |
| --- | --- |
| `admin_user_id` | Unique admin account ID |
| `email_address` | Admin login email |
| `password_hash` | Admin password hash |
| `full_name` | Admin full name |
| `is_active` | Whether admin account is active |
| `last_login_at` | Last login timestamp |
| `created_at` | Creation timestamp |
| `updated_at` | Update timestamp |

`site_settings`

| Column | Meaning |
| --- | --- |
| `setting_key` | Unique setting name |
| `setting_type` | Data type for validation: string, integer, decimal, boolean |
| `setting_value` | Stored setting value |
| `updated_at` | Last update timestamp |

Required initial keys include `whatsapp_number`, `premium_price_threshold`, and `currency`, and numeric settings must be parsed and validated by type before saving or use.[cite:53][cite:59][cite:73]

## Rule note

## Dynamic filters

Dynamic filters are implemented only through `filter_types`, `filter_values`, and `product_filter_values`, which means future filter expansion does not require schema changes to `products`.[cite:57][cite:61][cite:79]

### Review eligibility

A new review can be created only when a delivered `order_item` exists for the same customer and product, there is no existing review for that customer-product pair, and there is no matching row in `review_blocks`.[cite:57][cite:73]

If a cancellation is later approved for that order item, a `review_blocks` row is created if needed, which blocks only future review creation and does not remove any existing review.[cite:42][cite:45][cite:73]

### Shipping

If the selected address resolves to Gujarat through the normalized `states` lookup, shipping is free; otherwise the checkout service computes weight-based shipping using the active zone rate and minimum charge.[cite:43][cite:46][cite:49]

### Media upload

Admin product media must come from device file upload only, not pasted links, and the backend should enforce max media counts plus exactly one cover image per product before saving records.[cite:41][cite:44][cite:47]

### Implementation readiness note

The schema, constraints, and rule clarifications are now in a state that is ready for implementation planning, because the data model, validation expectations, and business-rule enforcement points are explicitly documented.[cite:73][cite:74][cite:79]